InterSystems
Creative data technology

Using Agentic AI with FHIR, Part One

Patrick W. Jamieson, M.D.
Technical Product Manager

Russ Leftwich, M.D.
Senior Clinical Advisor, Interoperability

Fan Ji,
Solution Developer

Agenda

01 Why Agentic AI + FHIR matters

02 Agentic AI Foundations

03 FHIR as the knowledge layer for agents

04 Terminology normalization and Code Resolution

05 Architecture preview: the FHIR Summarizer

Agenda

06 Patient retrieval with FHIR tools

07 From Bundle to compact clinical summary

08 Design patterns, safety, and production concerns

09 Q&A / Discussion

10 Want to Learn More

Want to Work with this Demo Software ?



- Clone or download the Open GitHub repository at: https://github.com/pjamiesointersystems/fhiragent.git
- Obtain a working copy of IRIS for Health Community Edition with a pre-configured FHIR repository at (Requires Docker): https://github.com/pjamiesointersystems/Dockerfhir.git
- Purchase an OpenAI key, for these demos we are using gpt-4.1-mini.

Confused About FHIR Installation?



Checkout the FHIR Environment Setup Guide

- https://community.intersystems.com/post/fhir-environment-setup-guide

- Video for Windows on YouTube at: https://www.youtube.com/watch?v=IyvuHbxCwCY

- Video for MacOS on YouTube at: https://www.youtube.com/watch?v=Ss7vU0l3JNU&t



01: Why Agentic AI + FHIR matters

The Healthcare Data Problem

Healthcare Data is Complex

Healthcare records are:

- Longitudinal
- Distributed across systems
- Highly structured but difficult to interpret
- Full of clinical terminology

FHIR makes this data **interoperable**, but not necessarily **easy to use**.

Why LLMs Alone Are Not Enough

Large Language Models Have Limitations

LLMs are powerful, but they cannot:

- Query FHIR servers
- Understand complex clinical graphs
- Maintain structured context across many resources
- Safely modify clinical data

To work with healthcare systems, AI must interact with **external tools and structured data**.

The Rise of Agentic AI

What Makes an AI System “Agentic”?

Agentic systems can:

- Plan tasks
- Call tools and APIs
- Retrieve external data
- Perform multi-step reasoning
- Iterate until a goal is achieved

This allows AI systems to **interact with real-world systems**, not just generate text.

The Opportunity: Agents + FHIR

FHIR provides:

- Structured healthcare data
- Standard APIs
- Clinical terminology integration
- Interoperability across systems

Agentic AI provides:

- Reasoning
- Tool orchestration
- Contextual understanding
- Clinical summarization

Together they enable **intelligent healthcare applications**.



02: Agentic AI Foundations

What Is a Large Language Model?

Understanding the Foundation

Key characteristics of LLMs:

- Trained on massive text corpora to predict next tokens
- Excel at reasoning, summarization, and language understanding
- Can follow complex instructions and maintain context
- Stateless by default: no memory between calls

LLMs are powerful reasoning engines, but they need tools to act on the world.

What Makes a System Agentic?

Characteristics of Agentic Systems

Agentic AI systems can:

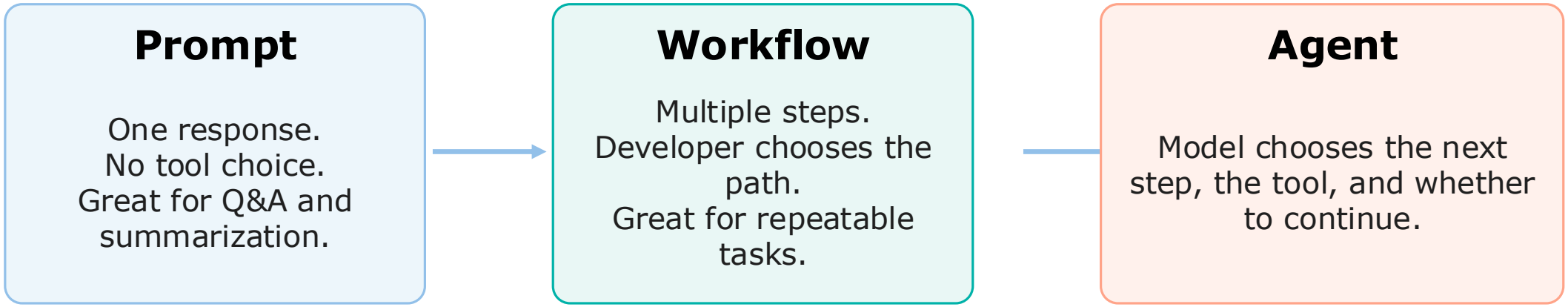
- Plan a sequence of actions
- Use external tools and APIs
- Retrieve and analyze structured data
- Maintain intermediate state
- Iterate until a goal is achieved



The LLM becomes the **reasoning engine**, not the entire system.

Prompt, Workflow, or Agent?

Three patterns — increasing flexibility, increasing control needs



FHIR can appear inside any of these patterns; in this workshop, it becomes part of the agent's tool and data layer.

Anatomy Of An Enterprise Agent



- User request
- Model / planner
- Tools and connectors
- State / memory
- Policy / approval layer
- Logs / evals / monitoring

When Should You Use an Agent?

Use agent behavior when flexibility matters — not just because the term is popular

✓ Good fit for an agent

- The task may take several steps
- The path is not fully known upfront
- More than one tool or system may be needed
- New information can change the plan
- A human can review important actions

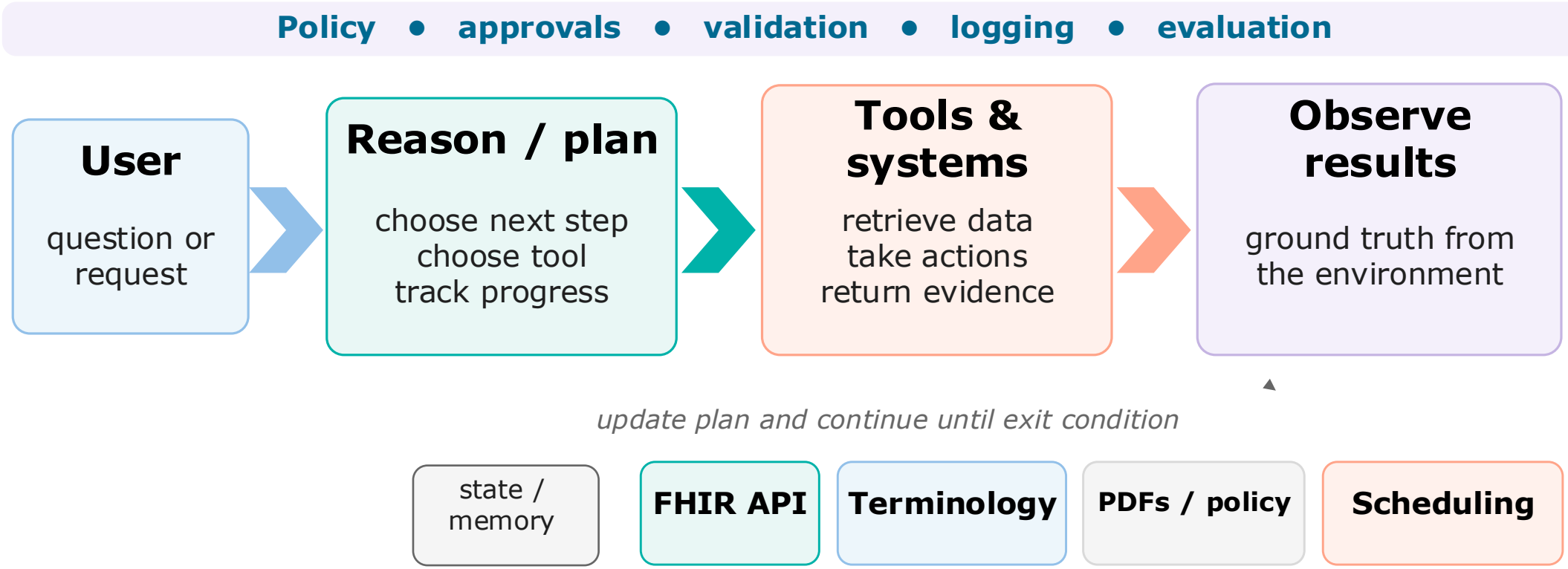
‖ Probably not an agent first

- It is only one lookup or one API call
- A deterministic business rule is enough
- The answer can be hardcoded or templated
- The cost of extra turns outweighs the benefit
- There is no approval path for high-risk actions

Start simple. Add agent autonomy only when it materially improves the outcome.

Anatomy of an Enterprise Agent

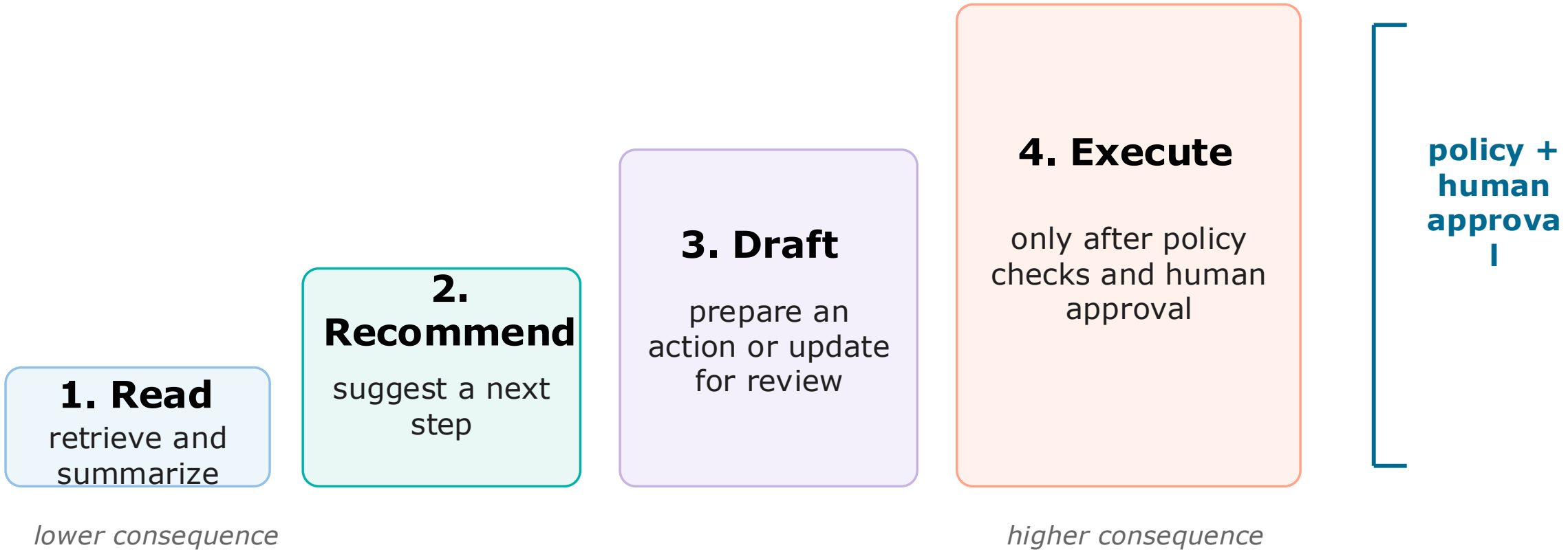
A model is only one part of the system



FHIR is a powerful healthcare-specific tool and knowledge layer inside this larger pattern.

Levels of Autonomy and Approval

Most real systems use graduated autonomy — not all-or-nothing automation



FHIR examples: read operations and write operations should not be treated the same way.

Common Failure Modes — and the Design Response

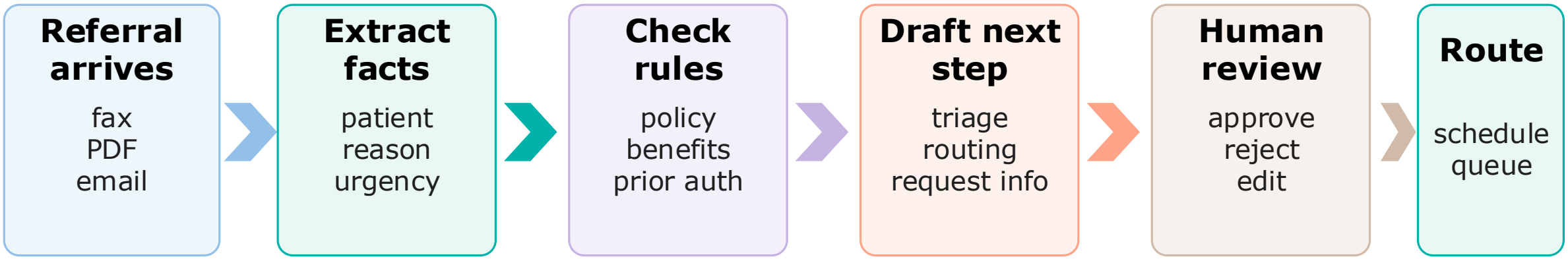
Reliable agents are engineered to fail safely and visibly

Wrong tool choice	clear tool descriptions + constrained tool access
Missing or partial data	source checks + evidence shown to the user
Looping or dead ends	max turns + retries + stop conditions
Ambiguous request	clarify, narrow scope, or hand off to a human
Unsafe or high-risk action	approval policy + audit log + least privilege

Production agents are systems with fallbacks, controls, and observability — not just prompts with ambition.

A Healthcare Example Without FHIR

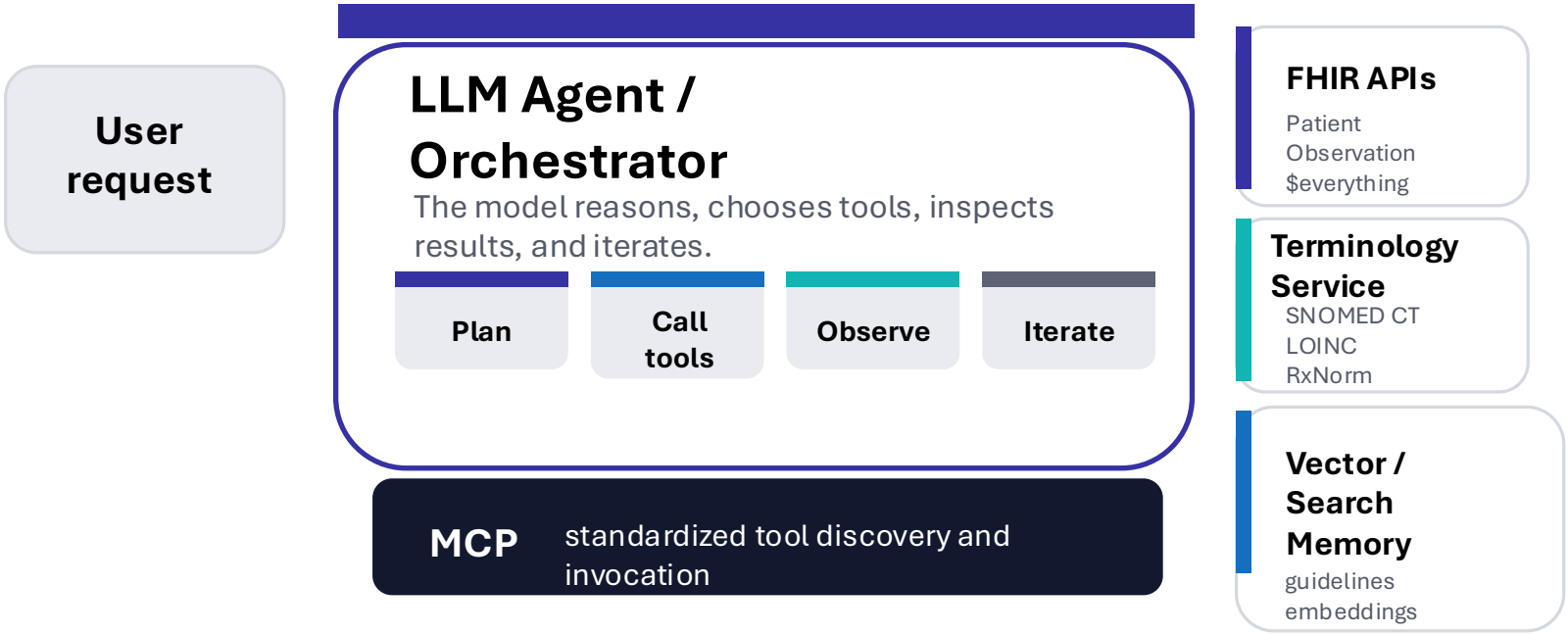
Referral intake



Same agent pattern: reason → use tools → observe → ask for approval when needed → act.

The Agent Architecture

A practical pattern: reasoning loop + standardized tool access + healthcare data services



Key point: the LLM is the reasoning layer; MCP standardizes tool access; FHIR and terminology services provide the healthcare data substrate.

The Tool-Using Agent Pattern

Agentic systems work by integrating tools.

Examples of tools:

- APIs
- Databases
- Search systems
- Vector indexes
- Terminology services
- FHIR servers



The agent decides which tool to call and when.

Multi-Step Reasoning



Agent Workflow

Agentic systems typically operate in a loop:

1. Understand the request
2. Decide which tools to use
3. Execute tool calls
4. Analyze results
5. Repeat if necessary
6. Generate final response

This pattern is often called **Reason** → **Act** → **Observe**.

Subagents and Specialized Roles

Agents Can Delegate Work

- Complex systems often use **specialized subagents**.

- Examples:

- Retrieval agents
- Terminology agents
- Summarization agents
- Workflow agents

Each agent focuses on a specific capability.



Model Context Protocol (MCP) At A Glance

HOST APP → MCP CLIENT → MCP SERVER → TOOLS / RESOURCES / PROMPTS

Claude Desktop IDE Enterprise chat UI

Connects the host to one or more servers

- capability negotiation
- session state
- message routing

Exposes capabilities through a standard interface

Tools actions the model can invoke

Resources structured or unstructured context

Prompts reusable workflows or templates

Think of MCP as a standard connector layer between an AI host and the external systems it needs to read from or act on.

In healthcare, those systems might include a FHIR repository, a terminology service, a policy engine, a vector store, or a scheduling system.

Who Built MCP — and Why It Emerged

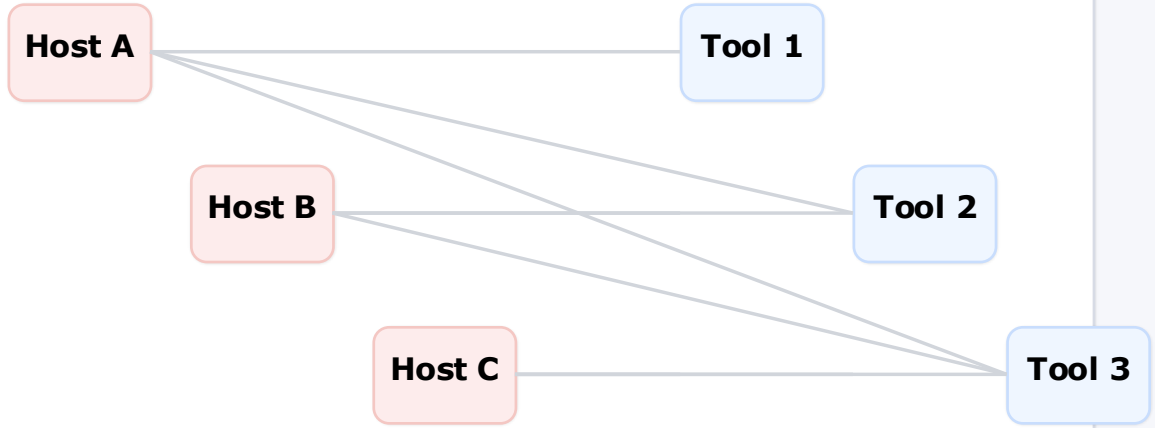
Who built it?

- Anthropic introduced and open-sourced MCP on Nov. 25, 2024.
- From the beginning, it was framed as an open standard rather than a product-specific API.
- The early release included the specification, SDKs, Claude Desktop support, and example servers for systems such as Google Drive, Slack, GitHub, Git, Postgres, and Puppeteer.

Why did it need to exist?

Before MCP

Every host × every tool required a custom integration.



With MCP

Hosts and tools implement one shared protocol, which reduces integration complexity and improves portability.

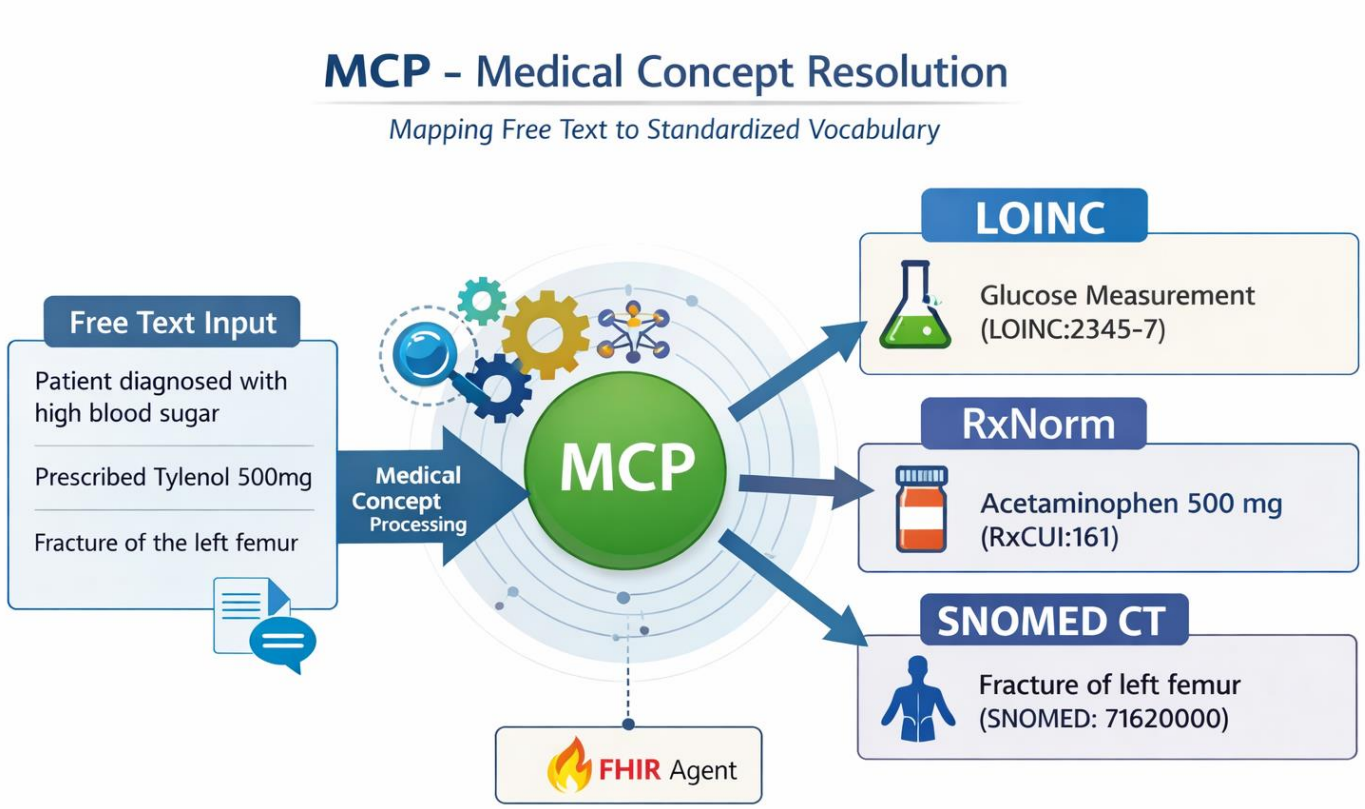
shared protocol

Why MCP Matters for Healthcare Agents

It gives you a cleaner way to compose specialized clinical systems



MCP does not replace FHIR. It gives agents a standard way to connect FHIR alongside the other systems a real healthcare workflow usually needs.


MCP - Medical Concept Resolution
Mapping Free Text to Standardized Vocabulary
Free Text Input
Patient diagnosed with high blood sugar
Prescribed Tylenol 500mg
Fracture of the left femur
Medical Concept Processing
MCP
FHIR Agent
LOINC
Glucose Measurement (LOINC:2345-7)
RxNorm
Acetaminophen 500 mg (RxCUI:161)
SNOMED CT
Fracture of left femur (SNOMED: 71620000)

What the Protocol Standardizes

The value is not just tool calling — it is the full connection contract

Lifecycle

initialize
capability
negotiation
shutdown

Transport

stdio for local servers
HTTP for remote servers
authorization support

Resources

readable context data
files, records, docs
structured or unstructured

Prompts

reusable interaction templates
workflows or starter instructions

Tools

callable actions
query APIs
run computations

- Hosts can connect to many servers without inventing a new connector model each time.
- Servers can expose capabilities in a way that is understandable to multiple host applications.
- Security, capability negotiation, and remote connectivity become part of the shared protocol conversation rather than ad hoc glue code.

Where MCP Appears to Be Headed

The trajectory is toward broader governance, discovery, and production readiness



Likely themes to watch

- better server discovery and packaging
- stronger trust, namespace verification, and security patterns
- more vendor-neutral governance and wider enterprise adoption

InterSystems
Creative data technology

03: FHIR as the Knowledge Layer for Agents



FHIR Resources and APIs

The Building Blocks

Core FHIR capabilities for agents:

- 150+ resource types: Patient, Condition, Observation, MedicationRequest
- RESTful API: GET, POST, PUT, DELETE on /Patient, /Observation, etc.
- Search parameters: _id, subject, code, date, _include, _revinclude
- Operations: $everything, $validate, $expand for bulk and specialized queries

FHIR provides a well-defined, queryable API that agents can call as tools.

FHIR Works with Agents

What FHIR Offers Agents:

- Well-defined RESTful API endpoints
- Standardized resource schemas (JSON/XML)
- Built-in search and filter parameters
- Terminology bindings and value sets

What Agents Add to FHIR:

- Natural language understanding of queries
- Multi-step retrieval and reasoning
- Contextual summarization of records
- Dynamic tool selection and orchestration

Together they create **an intelligent clinical data platform**.

Architecture Overview

How the Pieces Fit Together

Components of the agentic FHIR architecture:

- User query enters the agent orchestrator (LLM + tool loop)
- Terminology service normalizes codes and validates mappings via MCP
- Agent calls FHIR tools to retrieve patient data
- Bundle Extraction can pull out relevant FHIR structures
- LLM can use this in a nice summary if required

The agent orchestrates multiple services to transform a clinical question into a structured answer.

The FHIR Summarizer Pipeline

End-to-End Clinical Summary Generation

Four steps from raw data to narrative:

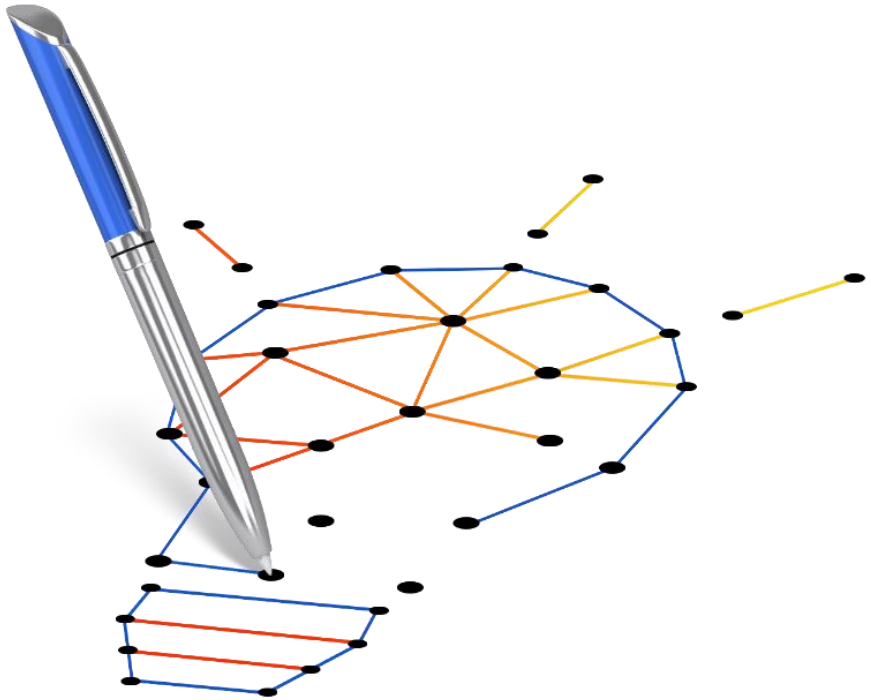
- Step 1: Retrieve patient record via $everything operation
- Step 2: Parse FHIR Bundle into structured clinical features
- Step 3: Extract conditions, medications, labs, encounters, procedures
- Step 4: Feed structured features to LLM for narrative summary



The pipeline transforms hundreds of FHIR resources into a concise, readable clinical summary.

Key Takeaways

FHIR as the Foundation for Agents



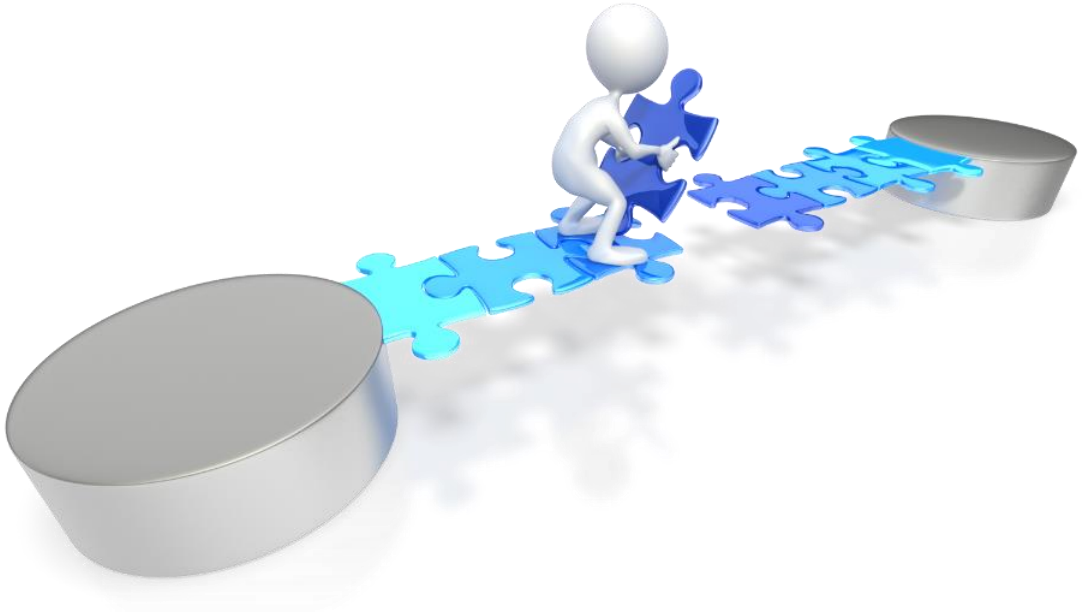
FHIR becomes the data layer for intelligent agents.

- FHIR is a **structured knowledge graph**
- Multiple retrieval patterns support different tasks
- $everything enables longitudinal analysis
- Raw FHIR must be **transformed before LLM use**

- Terminology codes provide **clinical meaning**

From Bundle to Clinical Summary

Extracting Structured Clinical Features

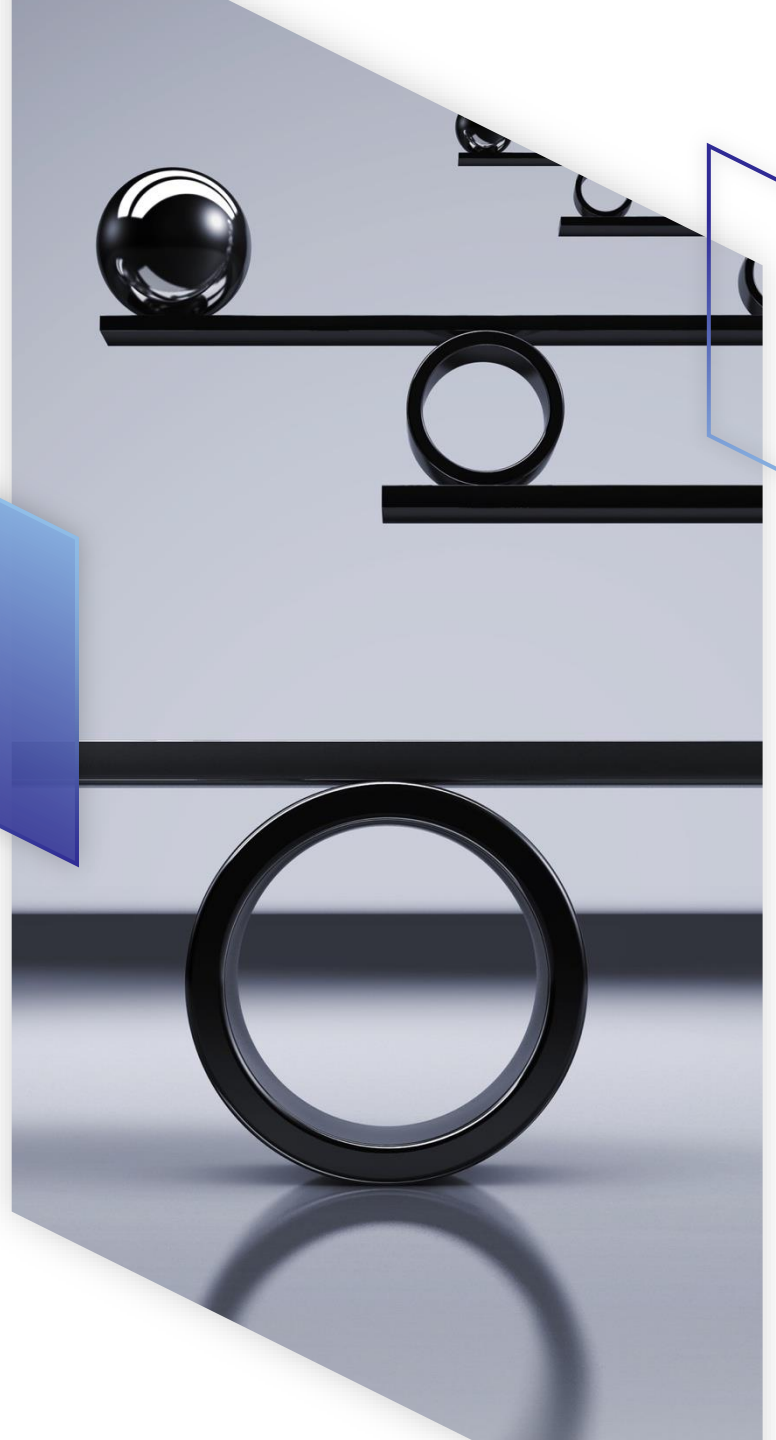


Structured feature extraction is the bridge between raw FHIR data and human-readable summaries.

- Transforming verbose FHIR into compact features:
  - Parse each resource type into domain-specific feature sets
  - Group conditions by active/resolved, medications by current/historical
  - Normalize lab values with units, reference ranges, and trends
  - Assemble features into a compact JSON structure for the LLM

InterSystems
Creative data technology

04: Terminology Normalization and Code Resolution



Terminology Normalization: The Hardest Problem in Clinical AI

Free text ≠ computable data

LLMs generate language, FHIR requires codes

Errors here silently break everything downstream

Why This is Hard

- Free text is ambiguous
- Multiple vocabularies
- Context matters

✓ medical-terminologies__snomed_search #call_AlC

SNOMED CT Search Results for "type 2 diabetes mellitus"

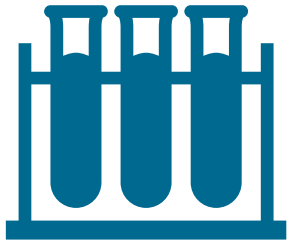
Found 5 concept(s):

SCTID	Preferred Term	FSN
44054006	Type 2 diabetes mellitus	Diabetes mellitus type 2 (disorder)
445353002	Brittle type 2 diabetes mellitus	Brittle type 2 diabetes mellitus (disorder)
721121000124104	Diabetes mellitus type 1 stage 2	Diabetes mellitus type 1 stage 2 (disorder)
81531005	Type 2 diabetes mellitus in obese	Diabetes mellitus type 2 in obese (disorder)
190389009	Type 2 diabetes mellitus with ulcer	Type II diabetes mellitus with ulcer (disorder)

Clinical Vocabularies



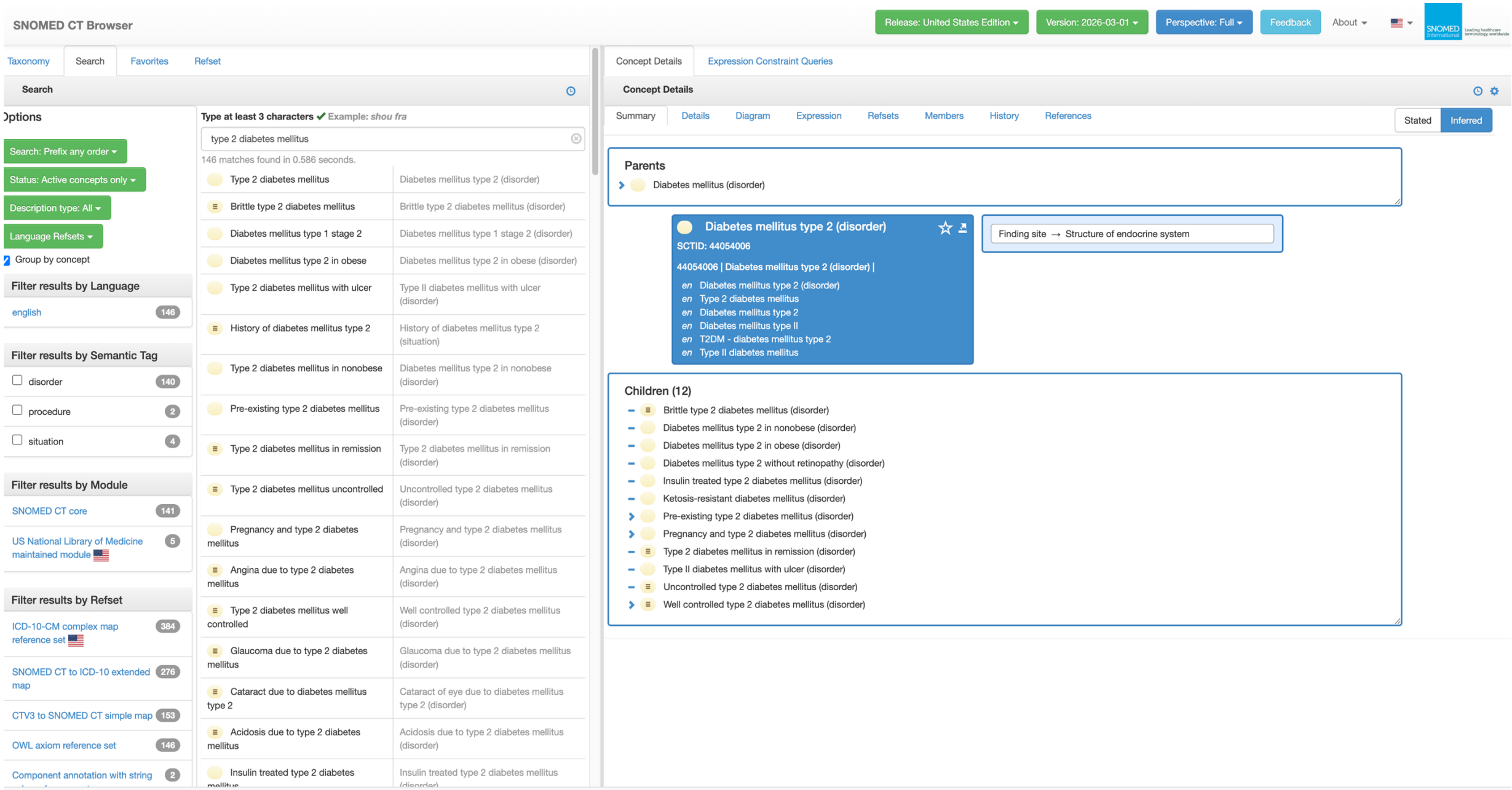
SNOMED CT – conditions



LOINC – labs



RxNorm – medications


SNOMED CT Browser
Release: United States Edition
Version: 2026-03-01
Perspective: Full
Feedback
About
Taxonomy
Search
Favorites
Refset
Search
Options
Search: Prefix any order
Status: Active concepts only
Description type: All
Language Refsets
Group by concept
Filter results by Language
english 146
Filter results by Semantic Tag
disorder 140
procedure 2
situation 4
Filter results by Module
SNOMED CT core 141
US National Library of Medicine maintained module 5
Filter results by Refset
ICD-10-CM complex map reference set 384
SNOMED CT to ICD-10 extended map 276
CTV3 to SNOMED CT simple map 153
OWL axiom reference set 146
Component annotation with string 2
Type at least 3 characters Example: shou fra
type 2 diabetes mellitus
146 matches found in 0.586 seconds.
Type 2 diabetes mellitus | Diabetes mellitus type 2 (disorder)
Brittle type 2 diabetes mellitus | Brittle type 2 diabetes mellitus (disorder)
Diabetes mellitus type 1 stage 2 | Diabetes mellitus type 1 stage 2 (disorder)
Diabetes mellitus type 2 in obese | Diabetes mellitus type 2 in obese (disorder)
Type 2 diabetes mellitus with ulcer | Type II diabetes mellitus with ulcer (disorder)
History of diabetes mellitus type 2 | History of diabetes mellitus type 2 (situation)
Type 2 diabetes mellitus in nonobese | Diabetes mellitus type 2 in nonobese (disorder)
Pre-existing type 2 diabetes mellitus | Pre-existing type 2 diabetes mellitus (disorder)
Type 2 diabetes mellitus in remission | Type 2 diabetes mellitus in remission (disorder)
Type 2 diabetes mellitus uncontrolled | Uncontrolled type 2 diabetes mellitus (disorder)
Pregnancy and type 2 diabetes mellitus | Pregnancy and type 2 diabetes mellitus (disorder)
Angina due to type 2 diabetes mellitus | Angina due to type 2 diabetes mellitus (disorder)
Type 2 diabetes mellitus well controlled | Well controlled type 2 diabetes mellitus (disorder)
Glaucoma due to type 2 diabetes mellitus | Glaucoma due to type 2 diabetes mellitus (disorder)
Cataract due to diabetes mellitus type 2 | Cataract of eye due to diabetes mellitus type 2 (disorder)
Acidosis due to type 2 diabetes mellitus | Acidosis due to type 2 diabetes mellitus (disorder)
Insulin treated type 2 diabetes mellitus | Insulin treated type 2 diabetes mellitus (disorder)
Concept Details
Expression Constraint Queries
Concept Details
Summary
Details
Diagram
Expression
Refsets
Members
History
References
Stated
Inferred
Parents
Diabetes mellitus (disorder)
Diabetes mellitus type 2 (disorder)
SCTID: 44054006
44054006 | Diabetes mellitus type 2 (disorder) |
en Diabetes mellitus type 2 (disorder)
en Type 2 diabetes mellitus
en Diabetes mellitus type 2
en Diabetes mellitus type II
en T2DM - diabetes mellitus type 2
en Type II diabetes mellitus
Finding site → Structure of endocrine system
Children (12)
Brittle type 2 diabetes mellitus (disorder)
Diabetes mellitus type 2 in nonobese (disorder)
Diabetes mellitus type 2 in obese (disorder)
Diabetes mellitus type 2 without retinopathy (disorder)
Insulin treated type 2 diabetes mellitus (disorder)
Ketosis-resistant diabetes mellitus (disorder)
Pre-existing type 2 diabetes mellitus (disorder)
Pregnancy and type 2 diabetes mellitus (disorder)
Type 2 diabetes mellitus in remission (disorder)
Type II diabetes mellitus with ulcer (disorder)
Uncontrolled type 2 diabetes mellitus (disorder)
Well controlled type 2 diabetes mellitus (disorder)

LOINC Hierarchy Browser — Patrick Jamieson

LOINCS: Hide | BRANCH NODES: Expand All, Collapse All | SEARCH: ystolic blood pressure, Search, Clear | HIERARCHY: Component By System | EXPORT

Category or Name	Component	Property	Timing	System	Scale	Method	Code
− {component} 109910							LP432695-7
− Laboratory 67299							LP29693-6
− HEDIS panels 198							LP71800-4
− HEDIS MY 2020 Value Sets 25							LP420644-9
− HEDIS MY 2020 Value Set - Systolic Blood Pressure 1							LP420642-3
− HEDIS MY 2020 Value Set - Systolic Blood Pressure | Patient | HEDIS panels 1							LP420740-5
HEDIS MY 2020 Systolic BP	HEDIS Measurement Year 2020 Value Set - S...	-	Pt	^Patient	-		96066-6
− HEDIS 2020 Value Sets 22							LP415726-1
− HEDIS 2020 Value Set - Systolic Blood Pressure 1							LP416925-8
− HEDIS 2020 Value Set - Systolic Blood Pressure | Patient | HEDIS panels 1							LP417327-6
HEDIS 2020 Systolic BP	HEDIS 2020 Value Set - Systolic Blood Press...	-	Pt	^Patient	-		94144-3
− Clinical 28782							LP7787-7
− Cardiology 1437							LP29708-2
− Cardiac ultrasound 615							LP29672-0
− Intravascular end systolic.XXX 1							LP72833-4
− Intravascular end systolic.XXX | Circulatory system XXX | Cardiac ultrasound 1							LP407930-9
BP end sys US	Intravascular end systolic.XXX	Pres	Pt	Circulatory system.XXX	Qn	US	20185-5
− Intravascular peak systolic.XXX 1							LP72835-9
− Intravascular peak systolic.XXX | Circulatory system XXX | Cardiac ultrasound 1							LP407931-7
BP peak sys US	Intravascular peak systolic.XXX	Pres	Pt	Circulatory system.XXX	Qn	US	20186-3
− Vital signs 605							LP30605-7
− Blood pressure 275							LP29704-1
− General blood pressure 7							LP7777-8
− Intravascular systolic 2							LP72836-7
− Intravascular systolic | Arterial system | General blood pressure 2							LP415811-1
BP sys	Intravascular systolic	Pres	Pt	Arterial system	Qn		8480-6
BP sys by Palpation	Intravascular systolic	Pres	Pt	Arterial system	Qn	Palpation	8479-8

National Library of Medicine
Lister Hill National Center for Biomedical Communications

RxNav — Navigating RxNorm Drugs

Name, NDC, RxCUI | lisinopril 10 MG

lisinopril 10 MG [RxCUI 316151]

Graph | RxCUI | NDC | RxTerms | Classes

Views
- Classic
- Simple
- Table

Filters
H V Rx S

Group | Form

Links

Legend
- MIN
- Pack
- Precise formulation
- Multi

Download

IN/MIN — Ingredient (2)
- H Rx S lisinopril
- M H Rx M hydroCHLOROthiazide / lisinopril

PIN — Precise Ingredient (0)

BN — Brand Name (3)
- H Rx S Prinivil
- H Rx M Zestoretic
- H Rx S Zestril

SCDC — Clinical Drug Component (1)
- H Rx SM lisinopril 10 MG

RxNav — Navigating RxNorm Drugs

SBDC — Branded Drug Component (3)
- H Rx S lisinopril 10 MG [Prinivil]
- H Rx S lisinopril 10 MG [Zestril]
- H Rx M hydroCHLOROthiazide 12.5 MG / lisinopril 10 MG [Zestoretic]

SCD/GPCK — Clinical Drug or Pack (2)
- H Rx S lisinopril 10 MG Oral Tablet
- H Rx M hydroCHLOROthiazide 12.5 MG / lisinopril 10 MG Oral Tablet

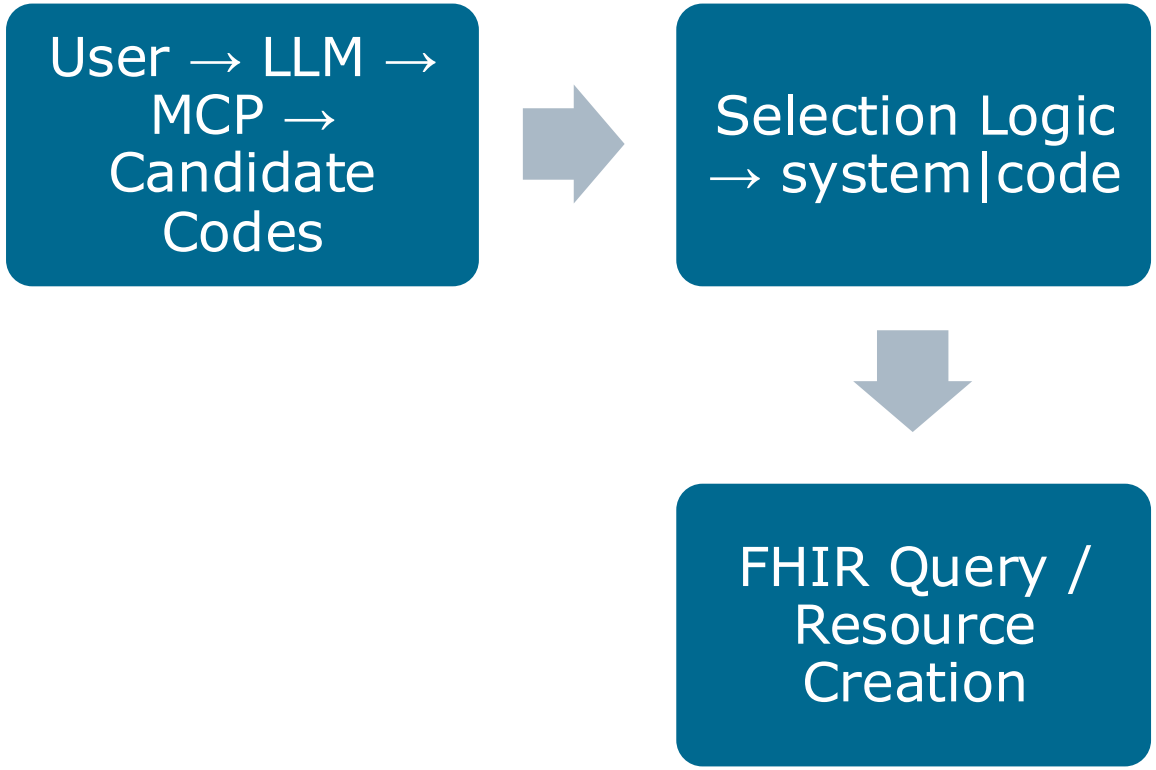
SBD/BPCK — Branded Drug or Pack (3)
- H Rx S Prinivil 10 MG Oral Tablet
- H Rx S Zestril 10 MG Oral Tablet, Once-Daily
- H Rx M Zestoretic 10 MG / 12.5 MG Oral Tablet

End-to-End Flow (Agent Reality)

Input: 'type 2 diabetes mellitus'

MCP returns 5+ SNOMED candidates

Agent must choose ONE



Bad vs Good Selection

❌ 'Type 2 diabetes with ulcer'

❌ 'Type 2 diabetes in obese'

✅ 'Type 2 diabetes mellitus (disorder)'

Selection = Intelligence Layer

Not solved by MCP

Requires reasoning over:

specificity	intent	clinical domain

Selection Heuristics (Production)

01

Prefer exact preferred term match

02

Avoid qualifiers unless explicitly stated

03

Map to correct domain (Condition vs Observation)

04

Fallback: top-ranked ONLY if safe

Solution – Terminology Service

User: “patients with diabetes”

Step 1: terminology service returns:

- Type 1
- Type 2
- Gestational
- General diabetes concept

Step 2: agent chooses broader concept OR asks clarification



Advanced: Context-Aware Resolution



'glucose' → LOINC (Observation)



'diabetes' → SNOMED (Condition)



'metformin' → RxNorm (MedicationRequest)

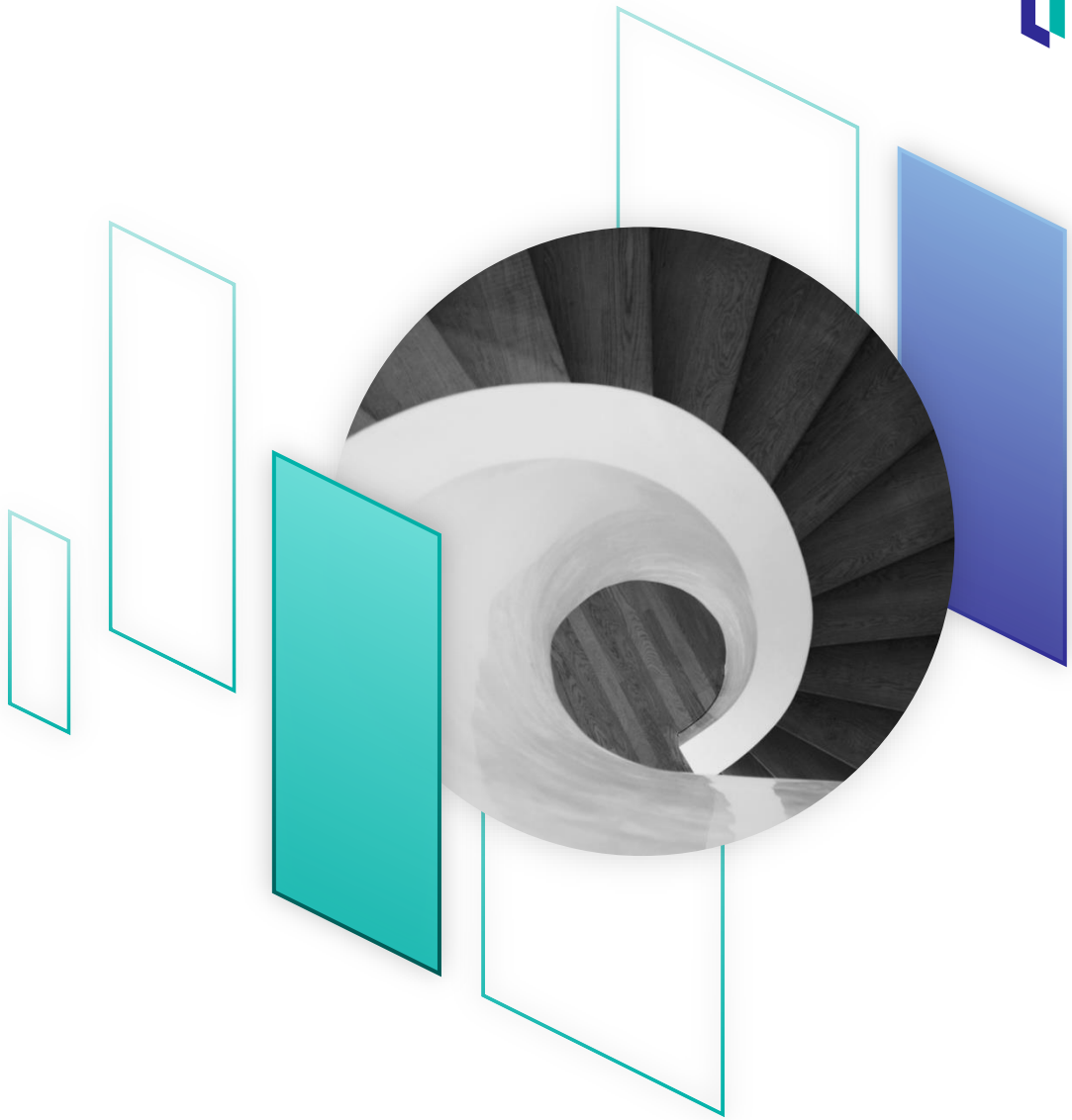
From Code → FHIR Token

Format: system|code

SNOMED:
http://snomed.info/sct|44054006

Used in search + resource coding

05: Architecture Preview – The FHIR Summarizer



Architecture Preview The FHIR Summarizer

Raw FHIR Bundles ≠ usable clinical insight

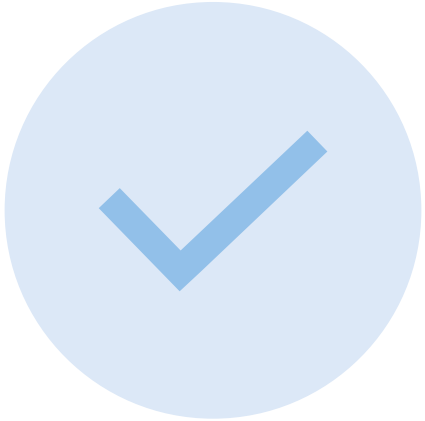
$everything → large, multi-resource payload

Need structured summarization layer

Why Subagents Matter



GENERAL AGENTS = TOO BROAD, TOO FRAGILE



SUBAGENT = SCOPED GOAL + CONSTRAINED TOOLS



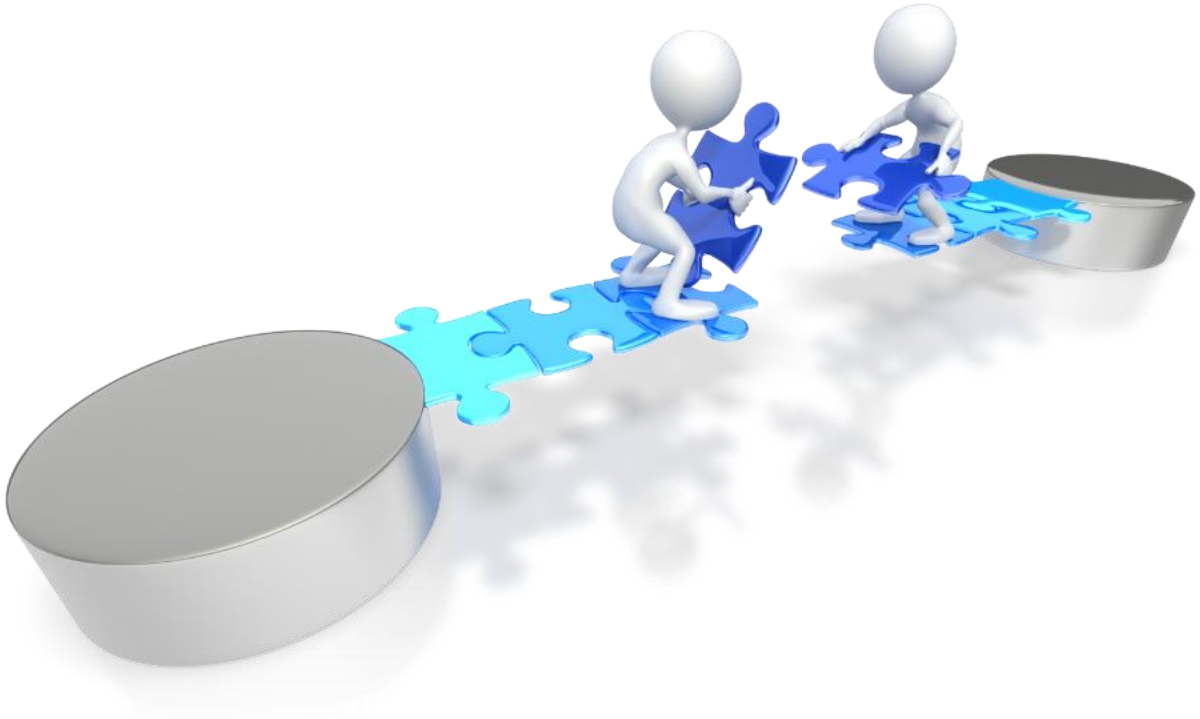
IMPROVES DETERMINISM AND REDUCES HALLUCINATION

The FHIR Summarizer Concept

A Specialized Subagent

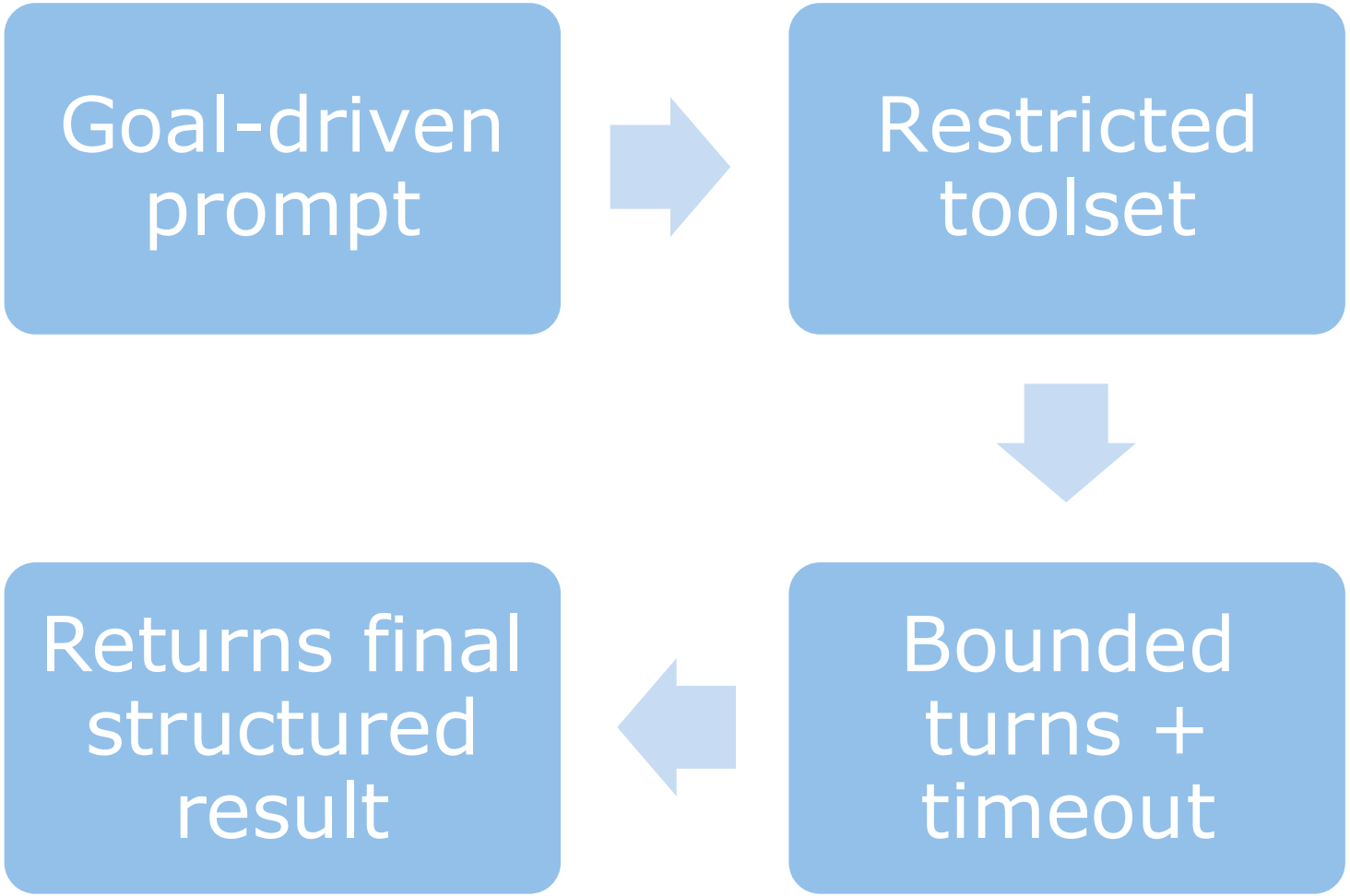
The FHIR Summarizer:

- Retrieves patient data
- Extracts key clinical features
- Reduces data size
- Produces structured summaries
- Feeds LLM for final output

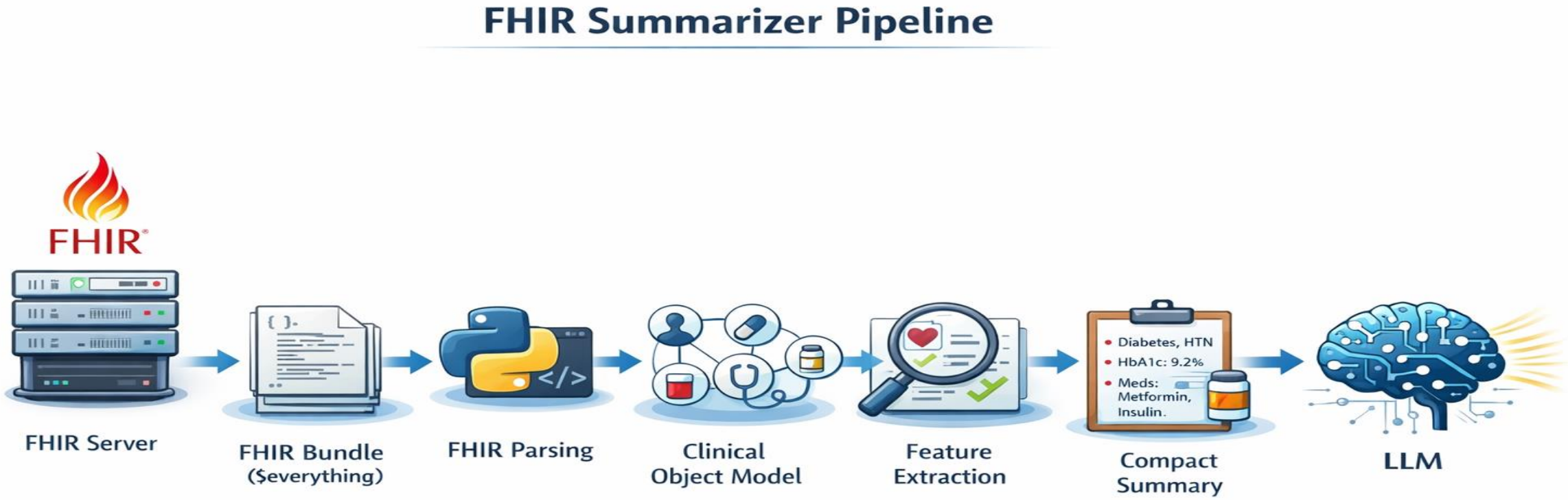


It acts as a **bridge between FHIR and AI reasoning**.

Subagent Execution Model



Architecture Overview



Step 1: Retrieval

Using FHIR Tools

Agent calls:

fhir_everything(patient_id="123")

Returns:

- Complete patient record
- Bundle of all related resources



Step 2: Parsing & Structuring

From JSON to Clinical Objects



Raw FHIR Bundle → Python objects

- Normalize structure
- Group related data
- Prepare for feature extraction

processor = FHIRBundleProcessor(bundle)

Step 3: Feature Extraction

Extract Clinically Relevant Data



We extract:

- Active conditions
- Recent lab results
- Medications
- Key clinical events

Result: A **focused clinical dataset**.

Step 4: Compact Summary

Reducing Token Load

Example Output

```
{
  "conditions": ["Diabetes", "Hypertension"],
  "labs": {"HbA1c": "9.2%"},
  "medications": ["Metformin", "Insulin"]
}
```

- Small
- Relevant
- LLM-ready

Tokens and Cost

- GPT-4.1-mini (as of April 2026) is priced at **$0.40 per 1 million input tokens** and **$1.60 per 1 million output tokens**. This model offers high cost-efficiency, featuring vision capabilities, 32,768 token output length, and prompt caching to further optimize costs for repeated prompts



Step 5: LLM Reasoning

Output:

- Clear narrative
- Clinically relevant
- Easy to interpret



PROMPT

SUMMARIZE THIS PATIENT'S CLINICAL STATUS:



{SUMMARY}

Clinical Reasoning Inside the Subagent

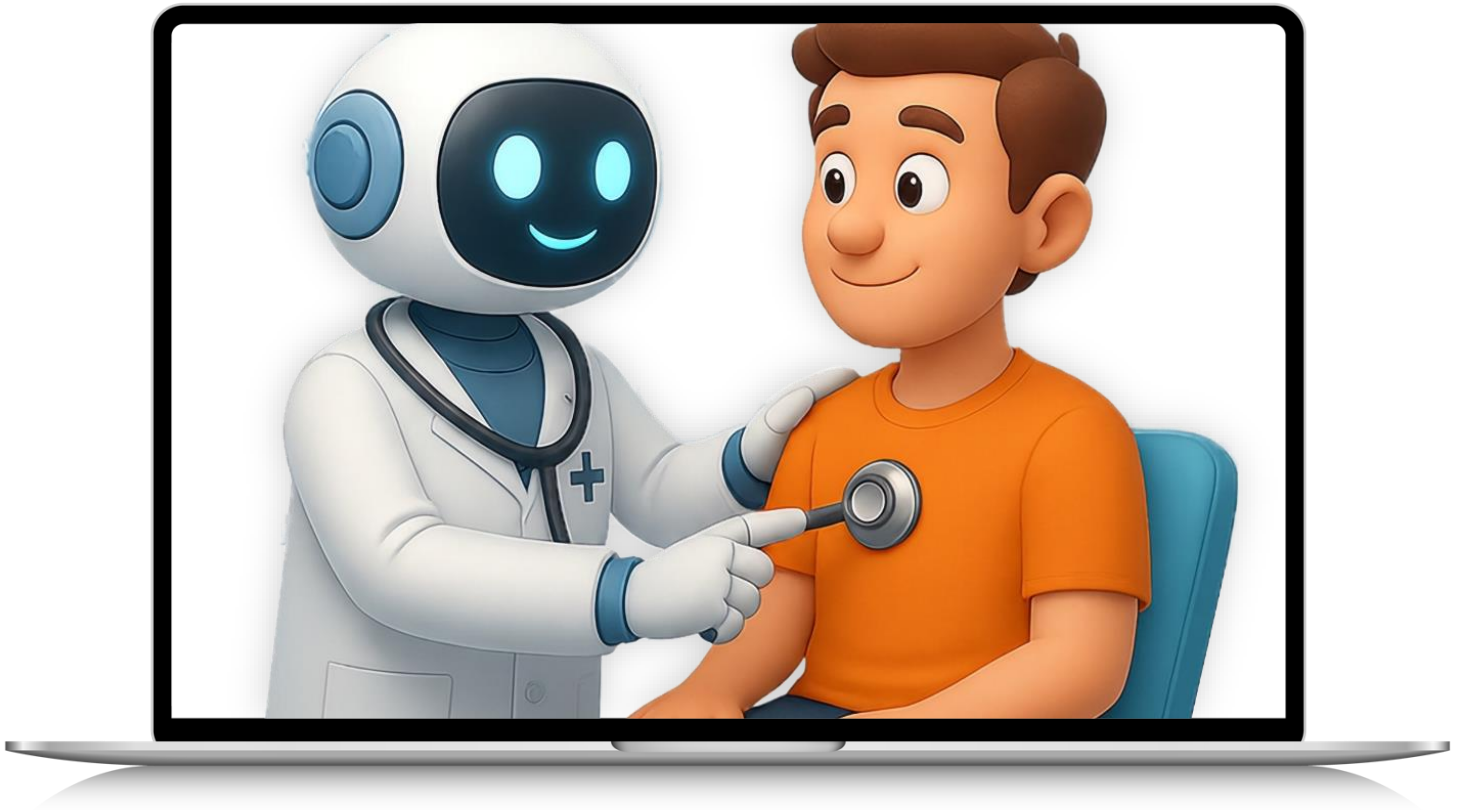
- Identify key resources (Condition, Observation, etc.)
- Extract clinically relevant fields
- Ignore noise (admin data)
- Produce concise structured output



AI-Agent

Open source tool for learning using AI Agents with FHIR

- Pure Python Client, does not use Claude Code
- Runs inside VSCode
- config.toml provides data such as LLM model name, FHIR Schema, and Base URL
- User Interface – console based with minor UI enhancements using Textual
- Not a Production System, easily extensible so you can add your own tools and subagents



https://github.com/pjamiesointersystems/fhiragent.git

MCP Server for Terminology Resolution

ICD-11, SNOMED CT, LOINC, RxNorm, MeSH

- Open Source
- https://github.com/SidneyBissoli/medical-terminologies-mcp.git

Tool	Description
snomed_search	Search concepts by term
snomed_concept	Details by SCTID
snomed_hierarchy	Parents/children (IS-A)
snomed_descriptions	FSN, PT, synonyms
snomed_ecl	ECL queries

Tool	Description
loinc_search	Search by term or code
loinc_details	Full code details
loinc_answers	Form answers list
loinc_panels	Related panels

Tool	Description
rxnorm_search	Search drugs by name
rxnorm_concept	Details by RxCUI
rxnorm_ingredients	Active ingredients
rxnorm_classes	Therapeutic classes
rxnorm_ndc	NDC mapping



Thank You